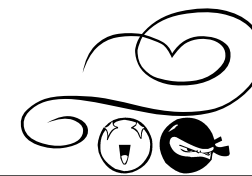


POLYNOMIALS

Part A: Terms



A1	DIFFICULTY: ☆	☒	☐
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What values of k make the polynomial $P(x) = (1-k^2)x^4 + (1+k)x^3 + x^2 + x + 1$ monic? 1

(A) $k = 0$ only

(C) $k = 0$ or $k = 1$

(B) $k = 0$ or $k = -1$

(D) $k = 0, k = -1, \text{ or } k = 1$

A2	DIFFICULTY: ☆☆	☒	☐
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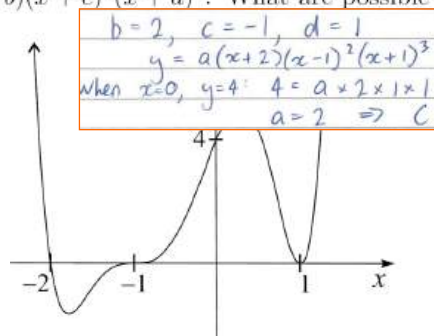
The diagram above shows the graph of $y = a(x+b)(x+c)^2(x+d)^3$. What are possible values of a, b, c and d ?

(A) $a = 1, b = 1, c = -2, d = -1$

(B) $a = 1, b = 2, c = -1, d = 1$

(C) $a = 2, b = 2, c = -1, d = 1$

(D) $a = 2, b = 2, c = 1, d = -1$



A3	DIFFICULTY: ☆	☒	☐
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Use division of polynomials to find the remainder $R(x)$, when

2

$P(x) = 4x^5 - 2x^3 + 2x^2 - 7$ is divided by $Q(x) = (2x^2 - 3)$.

$$\begin{array}{r}
 2x^3 + 2x + 1 \\
 2x^2 - 3 \overline{) 4x^5 - 2x^3 + 2x^2 - 7} \\
 \underline{4x^5 - 6x^3} \\
 4x^3 + 2x^2 - 7 \\
 \underline{4x^3 - 6x} \\
 2x^2 + 6x - 7 \\
 \underline{2x^2 - 3} \\
 6x - 4
 \end{array}$$

The remainder is $R(x) = (6x - 4)$

A4	DIFFICULTY: ☆	☒	☐
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A partially completed polynomial division is shown below.

1

$$\begin{array}{r}
 x^2 + x + 1 \\
 x^2 + 2x \overline{) x^4 + 3x^3 + 3x^2 - 4x + 5} \\
 \underline{x^4 + 2x^3} \\
 x^3 + 3x^2 \\
 \underline{x^3 + 2x^2} \\
 x^2 - 4x \\
 \underline{x^2 + 2x}
 \end{array}$$

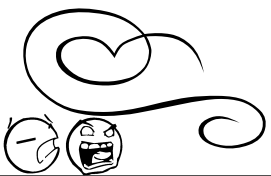
Which of the following expressions is equivalent to: $\frac{x^4 + 3x^3 + 3x^2 - 4x + 5}{x^2 + 2x}$?

A. $x^2 + x + 1 + \frac{5-6x}{x^2+2x}$

C. $x^2 + 2x + \frac{5-6x}{x^2+2x}$

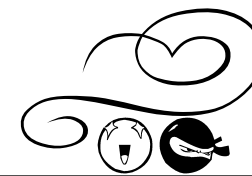
B. $x^2 + x + 1 + \frac{6x-5}{x^2+2x}$

D. $x^2 + 2x + \frac{6x-5}{x^2+2x}$



POLYNOMIALS

Part B: Factor Remainder



B1	DIFFICULTY: ☆☆	☒	☐
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A polynomial $P(x) = x^3 + ax^2 + bx + 6$ is divisible by both $(x + 3)$ and $(x + 1)$

- (a) Find the values of a and b . 2
 (b) Write the polynomial $P(x)$ as a product of its factors. 2

a) Factor theorem $P(-3) = (-3)^3 + a(-3)^2 + b(-3) + 6 = 0$ $-27 + 9a - 3b + 6 = 0$ $3a - b - 7 = 0$ ① $P(-1) = (-1)^3 + a(-1)^2 + b(-1) + 6 = 0$ $-1 + a - b + 6 = 0$ $a - b + 5 = 0$ ② Subtract equation (1) from equation (2). $2a - 12 = 0$ or $a = 6$ Substitute 6 for a into equation (2) $6 - b + 5 = 0$ or $b = 11$ $\therefore a = 6$ and $b = 11$	b) $P(x) = x^3 + 6x^2 + 11x + 6$ $= (x + 3)(x + 1)(x + \alpha)$ Required to find the value of α . By inspection of the constant term $(x + 2)$ is a factor. $\therefore P(x) = (x + 3)(x + 1)(x + 2)$
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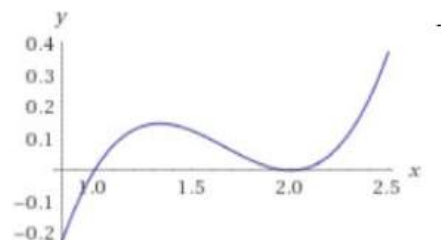
B2	DIFFICULTY: ☆☆	☒	☐
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- (i) Find the linear factors of $x^3 - 5x^2 + 8x - 4$ 2
 (ii) Hence solve $x^3 - 5x^2 + 8x - 4 > 0$ 2

(i) $x^3 - 5x^2 + 8x - 4 = (x - 2)^2(x - 1)$

Thus the linear factors are $(x - 2)$ and $(x - 1)$

- (ii) $x^3 - 5x^2 + 8x - 4 > 0$
 $1 < x < 2$ and $x > 2$



B3	DIFFICULTY: ☆	☒	☐
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The polynomial $P(x) = x^4 - kx^2 - 2x + 33$ has $(x - 3)$ as a factor. 1

What is the value of k ?

- (A) $-\frac{40}{3}$ (B) -12 (C) 12 (D) $\frac{40}{3}$

B4	DIFFICULTY: ☆☆☆	☒	☐
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A monic polynomial $P(x)$ of degree 4 has one zero equal to 2 and a double zero equal to -1 . If the polynomial passes through the point $(0, 6)$, Find the equation of $P(x)$. 3

Let α be the fourth zero of the given polynomial.

$$P(x) = (x - 2)(x + 1)^2(x - \alpha)$$

Since the given polynomial passes through $(0, 6)$

$$6 = (0 - 2)(0 + 1)^2(0 - \alpha)$$

$$6 = 2\alpha \quad \therefore P(x) = (x - 2)(x + 1)^2(x - 3)$$

$$\alpha = 3$$

B5	DIFFICULTY: ☆☆☆	☒	☐
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$P(x)$ has a remainder of 2 when divided by $(x - 3)$ and a remainder of -5 when divided by $(x - 2)$. 3

Find the remainder when $P(x)$ is divided by $x^2 - 5x + 6$.

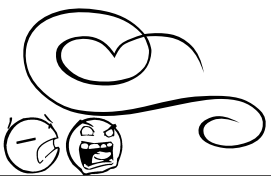
If $P(x)$ is divided by $x^2 - 5x + 6$, the degree of the remainder is at most 1. That is, linear in the form $Ax + B$.

$$\therefore P(x) = Q(x)(x^2 - 5x + 6) + (Ax + B)$$

Using the remainder theorem, we have $P(3) = 2$ and $P(2) = -5$.
 $\therefore 2 = 3A + B$ and $-5 = 2A + B$

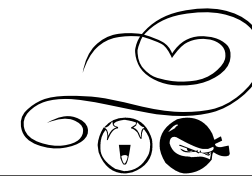
Solving simultaneously gives:
 $A = 7$ and $B = -19$

$$\therefore P(x) = Q(x)(x^2 - 5x + 6) + (7x - 19)$$



POLYNOMIALS

Part B: Factor Remainder



B6	DIFFICULTY: ☆☆	☒	☐
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A polynomial has a remainder of 1 when divided by $(x+1)$, and a remainder of 5 when divided by $(x-3)$. Find the remainder when the same polynomial is divided by $(x+1)(x-3)$.

2

$$\begin{aligned} \text{Let } P(x) &= (x+1)(x-3)Q(x) + R(x) \quad \text{where } R(x) = (Ax+B) \\ P(-1) &= 1 & P(3) &= 5 \\ -A+B &= 1 \dots (1) & 3A+B &= 5 \dots (2) \\ & & (2)-(1) & \\ & & 4A &= 4 \\ & & A &= 1 \text{ and } B = 2 \\ & \therefore R(x) &= x+2 \end{aligned}$$

B7	DIFFICULTY: ☆☆☆	☒	☐
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Let $P(x) = (x+1)(x-3)Q(x) + a(x+1) + b$, where $Q(x)$ is a polynomial and a and b are real numbers.

2

When $P(x)$ is divided by $(x+1)$ the remainder is -11 .

When $P(x)$ is divided by $(x-3)$ the remainder is 1.

(i) Find the value of b

(ii) What is the remainder when $P(x)$ is divided by $(x+1)(x-3)$?

$$\begin{aligned} \text{i) } P(-1) &= (0)Q(-1) + a(0) + b = -11 \\ \therefore b &= -11 \\ \text{ii) } P(3) &= (0)Q(3) + 4a + b = 1 \quad \left| \begin{array}{l} \text{when } P(x) \text{ is divided} \\ \text{by } (x+1)(x-3) \end{array} \right. \\ 4a + b &= 1 \\ 4a - 11 &= 1 \\ 4a &= 12 \\ a &= 3 \end{aligned}$$

$$\begin{aligned} \text{By } (x+1)(x-3) \quad R(x) &= a(x+1) + b \\ &= 3(x+1) - 11 \\ &= 3x + 3 - 11 \\ &= 3x - 8 \end{aligned}$$

B8	DIFFICULTY: ☆	☒	☐
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The polynomials $P(x)$ and $Q(x)$ are such that $P(x) = (x+3)(x-2)Q(x)$. When $P(x)$ is divided by $(x+1)$ the remainder is 12. What is the remainder when $Q(x)$ is divided by $(x+1)$?

1

- (A) -3 (B) -2 (C) 2 (D) 3

B9	DIFFICULTY: ☆	☒	☐
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Express $6x^3 - x^2 - 10x - 3$ as a product of three linear factors.

3

$$\begin{aligned} \text{Then } P(-1) &= 0 \Rightarrow (x+1) \text{ is a factor of } P(x). \\ 6x^3 - x^2 - 10x - 3 &= (x+1)(6x^2 - 7x - 3) \\ &= (x+1)(3x+1)(2x-3) \end{aligned}$$

B10	DIFFICULTY: ☆☆	☒	☐
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For what value of k does the polynomial $P(x) = x^2 + 5x + 4$ leave a remainder of k when divided by $(x-k)$?

1

- (A) $k = -2$ (C) $k = 1$
(B) $k = -1$ (D) $k = 2$

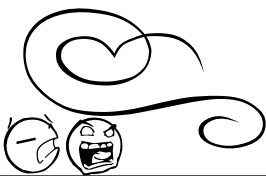
B11	DIFFICULTY: ☆	☒	☐
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If the polynomials $P(x) = 2x^3 + mx^2 + x - 3$ and $Q(x) = x^2 + 6x - 5$ have the same remainder when divided by $x+1$, find the value of m .

2

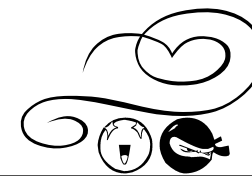
By Remainder Theorem,

$$\begin{aligned} Q(-1) &= (-1)^2 + 6(-1) - 5 \\ &= -10 \\ P(-1) &= 2(-1)^3 + m(-1)^2 + (-1) - 3 \\ &= -6 + m \\ \therefore -6 + m &= -10 \\ \therefore m &= -4 \end{aligned}$$



POLYNOMIALS

Part B: Factor Remainder



Unit

NAME: _____

B12 **DIFFICULTY: ☆☆**

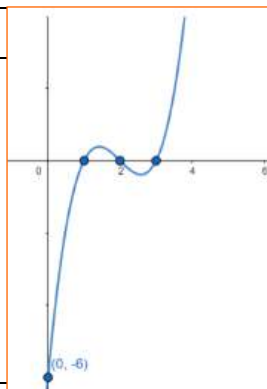
Consider the polynomial $P(x) = x^3 - 6x^2 + 11x - 6$.

- Solve the polynomial equation $P(x) = 0$.
- Hence or otherwise, sketch the function $y = P(x)$, labelling any x and y intercepts.

$$x = 1, x = 2 \text{ and } x = 3$$

x -intercepts at $x = 1, x = 2, x = 3$

y -intercept at -6



✖

2
1

B13 **DIFFICULTY: ☆**

Factorise $P(x) = 2x^3 + 3x^2 - 2x - 3$ completely.

$$\begin{array}{r}
 p(-1) = 0 \\
 \begin{array}{r}
 2x^2 + x - 3 \\
 (x+1) \overline{) 2x^3 + 3x^2 - 2x - 3} \\
 \underline{2x^3 + 2x^2} \\
 x^2 - 2x \\
 \underline{x^2 + x} \\
 -3x - 3 \\
 \underline{-3x - 3} \\
 0
 \end{array} \\
 p(x) = (x+1)(2x+3)(x-1)
 \end{array}$$

✓

✖

2

B14 **DIFFICULTY: ☆☆**

The polynomial $P(x)$ is given by $P(x) = x^3 + ax + b$ for real numbers a and b . $P(x) = 0$ has a root at $x = 2$. When $P(x)$ is divided by $(x+1)$ the remainder is -15 . Find the values of a and b .

$$\begin{array}{l}
 P(x) = x^3 + ax + b \\
 P(2) = 0 \Rightarrow 8 + 2a + b = 0 \quad \text{--- ①} \\
 P(-1) = -15 \Rightarrow -1 - a + b = -15 \quad \text{--- ②} \\
 \text{①} - \text{②}: \quad 9 + 3a = 15 \quad \text{into ①:} \\
 \quad \quad 3a = 6 \quad \quad 8 + 4 + b = 0 \\
 \quad \quad a = 2 \quad \quad b = -12
 \end{array}$$

B15 **DIFFICULTY: ☆☆**

Let the polynomial $P(x) = (x+1)(x-3)Q(x) + R(x)$, where $Q(x)$ and $R(x)$ are polynomials. When $P(x)$ is divided by $(x+1)$ the remainder is -7 .
 When $P(x)$ is divided by $(x-3)$ the remainder is 1 . Find $R(x)$.

$$P(x) = (x+1)(x-3)Q(x) + ax + b$$

$$P(-1) = -7$$

$$-7 = -a + b \quad (1)$$

$$P(3) = 1$$

$$1 = 3a + b \quad (2)$$

$$(1) - (2): \quad b = -7 + 2$$

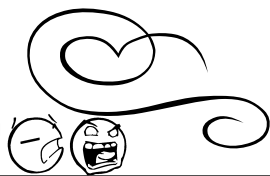
$$-8 = -4a \quad = -5$$

$$a = 2 \quad \therefore R(x) = 2x - 5$$

✓

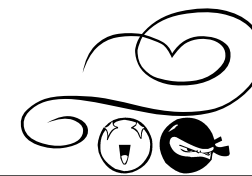
✖

3



POLYNOMIALS

Part C: Roots



C1	DIFFICULTY: ☆☆	☒	☐
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The equation $P(x) = x^3 - 6x^2 + 4x + 2$ has roots α, β and γ .

- (a) Find the value of $\alpha + \beta + \gamma$. 1
 (b) Find the value of $\alpha\beta\gamma$. 1
 (c) Find the value of $(\alpha - 3)(\beta - 3)(\gamma - 3)$. 1

(a)	$\alpha + \beta + \gamma = -\frac{b}{a} = -\frac{-6}{1} = 6$
(b)	$\alpha\beta\gamma = -\frac{d}{a} = -\frac{2}{1} = -2$
(c)	$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} = \frac{4}{1} = 4$ $(\alpha - 3)(\beta - 3)(\gamma - 3) = (\alpha\beta - 3\alpha - 3\beta + 9)(\gamma - 3)$ $= \alpha\beta\gamma - 3\alpha\gamma - 3\beta\gamma + 9\gamma - 3\alpha\beta + 9\alpha + 9\beta - 27$ $= \alpha\beta\gamma - 3(\alpha\beta + \beta\gamma + \alpha\gamma) + 9(\alpha + \beta + \gamma) - 27$ $= -2 - 3 \times 4 + 9 \times 6 - 27$ $= 13$

C2	DIFFICULTY: ☆☆	☒	☐
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Let the cubic polynomial $P(x) = x^3 - 3x^2 + kx + 12$ have roots α, β and γ .

- (i) Find the value of $\alpha + \beta + \gamma$ 1
 (ii) Find the value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$ 2
 (iii) Given that two of its roots have a sum of zero, find the value of k 2

(a) (i)	$\alpha + \beta + \gamma = 3$	(a) (iii)	Let the roots be $\alpha, -\alpha$ and β
(a) (ii)	$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma}$ $= \frac{3}{-12}$ $= -\frac{1}{4}$	$\alpha - \alpha + \beta = 3$ $\beta = 3$ $(3)^3 - 3(3)^2 + 3k + 12 = 0$ $3k = -12$ $k = -4$	$P(3) = 0$ $3k = -12$ $k = -4$

C3	DIFFICULTY: ☆	☒	☐
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One of the roots of the equation $x^3 + kx^2 + 1 = 0$ is the sum of the other two roots.

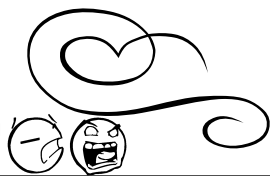
- (i) Show that $x = -\frac{k}{2}$ is a root of the equation. 2
 (ii) Find the value of k . 2

(i)	Let roots be $\alpha, \beta, \alpha + \beta$. Sum of roots $= \alpha + \beta + (\alpha + \beta) = 2\alpha + 2\beta = -k$ $\therefore \alpha + \beta = -\frac{k}{2}$ Since $\alpha + \beta$ is one of the roots, $-\frac{k}{2}$ is one of the roots.	(ii)	Product of roots $\alpha\beta(\alpha + \beta) = -1$ $\alpha\beta(-\frac{k}{2}) = -1$ $\alpha\beta = \frac{2}{k}$ Sum of pairs $\alpha\beta + \alpha(\alpha + \beta) + \beta(\alpha + \beta) = 0$ $\frac{2}{k} + (\alpha + \beta)^2 = 0$ $\frac{2}{k} + (-\frac{k}{2})^2 = 0$ $\frac{k^2}{4} = -\frac{2}{k}$ $k^3 = -8$ $k = -2$
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C4	DIFFICULTY: ☆	☒	☐
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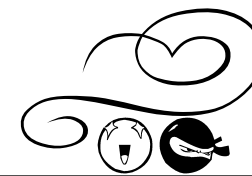
If α, β and γ are the roots of $3x^3 + 4x^2 - 5x - 8 = 0$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$ $= \frac{-5/3}{8/3} = -5/8$
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POLYNOMIALS

Part C: Roots



C5	DIFFICULTY: ☆☆☆	☒	☐
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A polynomial of degree 4 has a double zero at 2 with the other two zeros being k and $-k$.

When $x = 1$ it takes the value 8 and when $x = 3$ it takes the value 80.

Find the possible values of k .

3

$$\begin{aligned}
 P(x) &= a(x-2)^2(x-k)(x+k) \\
 P(1) &= a(1-2)^2(1-k)(1+k) = 8 & \therefore \textcircled{2} \div \textcircled{1} \frac{a(9-k^2)}{a(1-k^2)} &= 10 \\
 \therefore a(1-k^2) &= 8 - \textcircled{1} & \therefore (9-k^2) &= 10(1-k^2) \\
 P(3) &= a(3-2)^2(3-k)(3+k) = 80 & \therefore 9-k^2 &= 10-10k^2 \\
 \therefore a(9-k^2) &= 80 - \textcircled{2} & \therefore 9k^2 &= 1, k = \pm \frac{1}{3}
 \end{aligned}$$

C6	DIFFICULTY: ☆☆☆	☒	☐
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If α, β and γ are the roots of the cubic equation $2x^3 + 4x - 3 = 0$.

Find the value of $\alpha^3 + \beta^3 + \gamma^3$.

$$\begin{aligned}
 2x^3 + 4x - 3 &= 0 \therefore 2\alpha^3 + 4\alpha - 3 = 0 \\
 \therefore 2\alpha^3 &= -4\alpha + 3 \text{ and similar for } \beta^3 \text{ and } \gamma^3 \\
 \therefore \alpha^3 + \beta^3 + \gamma^3 &= \\
 (-4\alpha + 3 - 4\beta + 3 - 4\gamma + 3)/2 &= \\
 = (-4(\alpha + \beta + \gamma) + 9)/2 &= (-4(0) + 9)/2 = 9/2
 \end{aligned}$$

3

C7	DIFFICULTY: ☆☆	☒	☐
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Consider the polynomial $P(x) = x^3 - px^2 + qx - pq$, where $q > 0$.

(i) Show that $(x-p)$ is a factor of $P(x)$.

(ii) Show that the equation $P(x) = 0$ has only one real root.

1

2

$$\begin{aligned}
 \text{(i)} \quad P(p) &= p^3 - pp^2 + qp - pq = 0 & \text{(ii)} \quad P(x) &= x^2(x-p) + q(x-p) \\
 \therefore (x-p) &\text{ is a factor of } P(x). & &= (x-p)(x^2 + q) = 0 \\
 & & &x^2 = -q < 0 \rightarrow \text{no solution (since } q > 0) \\
 & & &\therefore x = p \text{ only.}
 \end{aligned}$$

C8	DIFFICULTY: ☆☆	☒	☐
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The polynomial $P(x) = x^3 + px^2 + qx + r$ has roots 0, 2, -2 .

i) Find the values of p, q and r .

2

ii) Sketch the graph of $y = P(x)$, showing any intercepts with the coordinate axes.

2

iii) Hence, or otherwise, solve $\frac{x^2-4}{x} \geq 0$

2

i) $P(0) = 0$
 $0^3 + p(0)^2 + q(0) + r = 0$
 $\therefore r = 0$
 $P(2) = 0$
 $(2)^3 + p(2)^2 + q(2) + 0 = 0$
 $8 + 4p + 2q = 0 \dots\dots\dots(1)$
 $P(-2) = 0$
 $(-2)^3 + p(-2)^2 + q(-2) + 0 = 0$
 $-8 + 4p - 2q = 0 \dots\dots\dots(2)$
 $\text{eqn (1) + (2)} \quad 0 + 8p + 0 = 0$
 $\therefore p = 0$
 $\text{Sub } p \text{ into (2): } -8 + 4(0) - 2q = 0$
 $\therefore q = -4$

ii) $P(x) = x^3 - 4x$
 $= x(x^2 - 4)$
 $= x(x-2)(x+2)$

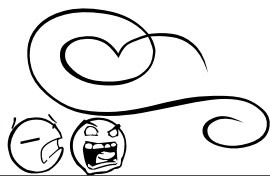
iii) $\frac{x^2-4}{x} \geq 0 \quad x \neq 0$
 $x^2 \times \frac{(x^2-4)}{x} \geq 0$
 $x(x-2)(x+2) \geq 0$
 $\text{From part (ii) } x(x-2)(x+2) \geq 0 \text{ when}$
 $-2 \leq x < 0 \text{ and } x \geq 2$

C9	DIFFICULTY: ☆☆	☒	☐
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The zeros of $x^3 - 5x + 3 = 0$ are α, β and γ . Without evaluating α, β and γ , find a cubic equation with integer coefficients whose zeros are $2\alpha, 2\beta$ and 2γ .

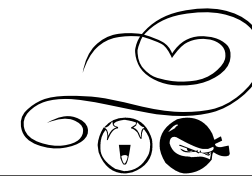
2

$$\begin{aligned}
 \alpha + \beta + \gamma &= \frac{-b}{a} & \therefore 2\alpha^2\beta + 2\alpha^2\gamma + 2\beta^2\gamma &= 4(\alpha\beta + \alpha\gamma + \beta\gamma) \\
 &= 0 & &= -20 \\
 \therefore 2\alpha + 2\beta + 2\gamma &= 2(\alpha + \beta + \gamma) & \alpha\beta\gamma &= \frac{-d}{a} \\
 &= 0 & &= -3 \\
 \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} & 2\alpha^2\beta^2\gamma &= 8\alpha\beta\gamma \\
 &= -5 & &= -24 \\
 \therefore \text{New cubic equation with integer coefficients is } & & &x^3 - 20x + 24 = 0
 \end{aligned}$$



POLYNOMIALS

Part C: Roots



C10	DIFFICULTY: ☆☆	☒	☐
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If α , β and γ are the roots of $2x^3 + 3x^2 - 5x + 1 = 0$ find:

- (i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2
- (ii) $\alpha^2 + \beta^2 + \gamma^2$ 2
- (iii) $(\alpha\beta)^2\gamma + \alpha(\beta\gamma)^2 + (\alpha\gamma)^2\beta$ 2

i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

$$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{-5/2}{-1/2} = 5$$

iii) $(\alpha\beta)^2\gamma + \alpha(\beta\gamma)^2 + (\alpha\gamma)^2\beta$

$$= \alpha\beta\gamma(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= \left(-\frac{1}{2}\right)\left(-\frac{5}{2}\right) = \frac{5}{4}$$

ii) $\alpha^2 + \beta^2 + \gamma^2$

$$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= \left(-\frac{3}{2}\right)^2 - 2\left(-\frac{5}{2}\right) = \frac{29}{4} = 7\frac{1}{4}$$

C11	DIFFICULTY: ☆☆	☒	☐
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The equation $x^3 - 8x^2 + 4x + 2 = 0$ has roots α , β and γ .

- (i) Find the values of $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$. 2
- (ii) Hence find the value of $\frac{1+\alpha}{\alpha\beta} + \frac{1+\beta}{\beta\gamma} + \frac{1+\gamma}{\gamma\alpha}$ 2

$\alpha + \beta + \gamma = 8, \alpha\beta + \beta\gamma + \gamma\alpha = 4, \alpha\beta\gamma = -2$

ii. Outcomes assessed: ME11-2

$$\frac{1+\alpha}{\alpha\beta} + \frac{1+\beta}{\beta\gamma} + \frac{1+\gamma}{\gamma\alpha} = \frac{\gamma(1+\alpha) + \alpha(1+\beta) + \beta(1+\gamma)}{\alpha\beta\gamma}$$

$$= \frac{(\alpha + \beta + \gamma) + (\alpha\beta + \beta\gamma + \gamma\alpha)}{\alpha\beta\gamma} = \frac{8+4}{-2} = -6$$

C12	DIFFICULTY: ☆	☒	☐
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For what value of k does the polynomial equation $x^3 - 3x^2 - 6x + 8 = 0$ have roots of the form $\frac{k}{r}$, k and kr for some constant r ? 1

- (A) $k = -2$ (C) $k = 1$
- (B) $k = -1$ (D) $k = 2$

C13	DIFFICULTY: ☆	☒	☐
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For what value of k does the polynomial equation $x^3 + 3x^2 - 6x - 8 = 0$ have roots of the form $k-d$, k and $k+d$ for some constant d ? 1

- (A) $k = -2$ (C) $k = 1$
- (B) $k = -1$ (D) $k = 2$

C14	DIFFICULTY: ☆	☒	☐
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For what value of k is the polynomial $P(x) = kx^2 + (2k-1)x + k$ a perfect square? 1

- (A) $k = \frac{1}{9}$ (C) $k = 4$
- (B) $k = \frac{1}{4}$ (D) $k = 9$

$$P(x) = k\left\{x^2 + \left(2 - \frac{1}{k}\right)x + 1\right\} \therefore k \text{ is a perfect square and } 2 - \frac{1}{k} = -2$$

$$k = \frac{1}{4}$$

C15	DIFFICULTY: ☆☆	☒	☐
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The cubic equation $x^3 + (p-1)x^2 + (q+2)x + 4 = 0$ has 3 non-zero roots α, β and γ . 2

Find in terms of the constant p and q the value of $\alpha^2 + \beta^2 + \gamma^2$.

(Leave your answer in its simplest form.)

$$x^3 + (p-1)x^2 + (q+2)x + 4 = 0$$

Let the roots be α, β and γ .

$$\alpha + \beta + \gamma = -(p-1) = 1-p$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = q+2$$

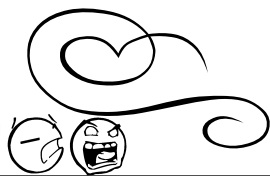
$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= [-(p-1)]^2 - 2(q+2)$$

$$= (p-1)^2 - 2(q+2)$$

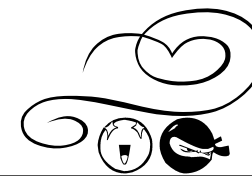
$$= p^2 - 2p + 1 - 2q - 4$$

$$= p^2 - 2p - 2q - 3 \rightarrow 1$$



POLYNOMIALS

Part C: Roots



C16	DIFFICULTY: ☆☆	☒	☐
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If the polynomial, $x^3 - x + 2 = 0$, has roots α, β and γ , show that $\alpha\beta$ is one of the roots of the polynomial $x^3 + x^2 - 4 = 0$. 2

Since α, β and γ are roots of the given polynomial, $x^3 - x + 2 = 0$, we consider the sum of roots 3 at a time:
 $\alpha\beta\gamma = -2$

$$\therefore \alpha\beta = \frac{-2}{\gamma}$$

Substitute into the second equation given, i.e. $P(x) = x^3 + x^2 - 4$, to check whether $\alpha\beta$ is a root and hence if $P(\alpha\beta) = 0$.

$$P(\alpha\beta) = P\left(\frac{-2}{\gamma}\right) = \left(\frac{-2}{\gamma}\right)^3 + \left(\frac{-2}{\gamma}\right)^2 - 4$$

Rearrange to give:

$$= \frac{-4}{\gamma^3}(\gamma^3 - \gamma + 2)$$

$$= \frac{-4}{\gamma^3} \times 0 = 0$$

Since γ is a root to the first equation, i.e. $\gamma^3 - \gamma + 2 = 0$

Therefore $\alpha\beta$ is one of the roots of the polynomial $x^3 + x^2 - 4 = 0$.

C17	DIFFICULTY: ☆☆	☒	☐
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In the quadratic equation, $x^2 + 3x + 2p - 1 = 0$, one root is triple the other. 2

Find the value of p .

Let the roots be α & 3α

$$\Rightarrow 3\alpha^2 = 2p - 1 \quad (\text{Prod of roots}) \quad \text{--- ①}$$

$$\text{and } 4\alpha = -3 \quad (\text{Sum of roots}) \quad \text{--- ②}$$

$$\text{② } \Rightarrow \alpha = -\frac{3}{4}$$

in ① $3\left(-\frac{3}{4}\right)^2 = 2p - 1$

$$\Rightarrow 2p - 1 = \frac{27}{16}$$

$$\therefore p = \frac{43}{32}$$

C18	DIFFICULTY: ☆☆	☒	☐
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If α, β and γ are the roots of the cubic equation 2

$5x^3 + 2x^2 - x - 3 = 0$, find the value of $\alpha^2 + \beta^2 + \gamma^2$

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) \\ &= \left(-\frac{2}{5}\right)^2 - 2\left(\frac{-1}{5}\right) = \frac{14}{25} \end{aligned}$$

C19	DIFFICULTY: ☆☆	☒	☐
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The polynomial $G(x) = 2x^3 - 3x^2 + 7x - 5$ has roots α, β and γ . 2

Find the value of $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$.

$$\begin{aligned} \alpha + \beta + \gamma &= -\frac{b}{a} = \frac{3}{2} \\ \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} = \frac{7}{2} \\ \alpha\beta\gamma &= -\frac{d}{a} = -\frac{5}{2} \\ \alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 &= \alpha\beta\gamma(\alpha + \beta + \gamma) \\ &= \frac{5}{2} \times \frac{3}{2} = \frac{15}{4} \end{aligned}$$

C20	DIFFICULTY: ☆☆	☒	☐
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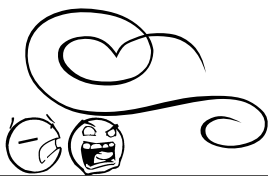
The cubic equation $2x^3 - 3x + 2 = 0$ has three real roots α, β and γ .

Evaluate:

(i) $\alpha^3\beta^3\gamma^2 + \alpha^3\beta^2\gamma^3 + \alpha^2\beta^3\gamma^3$ 2

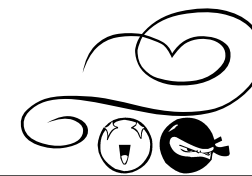
(ii) $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta}$

$$\begin{aligned} \alpha + \beta + \gamma &= 0 & \text{(ii) } \frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta} \\ \alpha\beta + \beta\gamma + \gamma\alpha &= -\frac{3}{2} & = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma} \\ \alpha\beta\gamma &= -1 & = \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)}{\alpha\beta\gamma} \\ \text{(i) } \alpha^3\beta^3\gamma^2 + \alpha^3\beta^2\gamma^3 + \alpha^2\beta^3\gamma^3 & & = \frac{0^2 - 2 \times -\frac{3}{2}}{-1} = -3 \\ &= \alpha^2\beta^2\gamma^2(\alpha + \beta + \gamma) & \\ &= (-1)^2 \left(-\frac{3}{2}\right) = -\frac{3}{2} & \end{aligned}$$



POLYNOMIALS

Part C: Roots



C21

DIFFICULTY: ☆☆



The cubic equation $ax^3 + bx^2 + cx + d = 0$ has one root which is equal to the average of the other two roots. Show that $27a^2d + 2b^3 = 9abc$.

2

$$(d) \quad ax^3 + bx^2 + cx + d = 0$$

Let the roots be 2α , 2β , $\alpha + \beta$.

$$2\alpha + 2\beta + \alpha + \beta = -\frac{b}{a} \quad \text{sum of roots}$$

$$3(\alpha + \beta) = -\frac{b}{a}$$

$$\alpha + \beta = -\frac{b}{3a} \quad \checkmark$$

Note: $\frac{2\alpha + 2\beta}{2} = \alpha + \beta$,
so $\alpha + \beta$ is the
average of 2α , 2β .

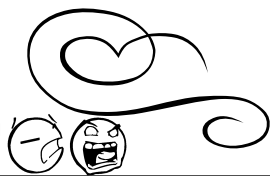
But $\alpha + \beta$ is a root, so we can
sub $x = -\frac{b}{3a}$ into the cubic equation:

$$a\left(-\frac{b}{3a}\right)^3 + b\left(-\frac{b}{3a}\right)^2 + c\left(-\frac{b}{3a}\right) + d = 0$$

$$\frac{-ab^3}{27a^3} + \frac{b^3}{9a^2} - \frac{bc}{3a} + d = 0$$

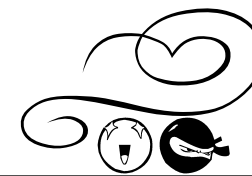
$$-b^3 + 3b^3 - 9abc + 9a^2d = 0$$

$$27a^2d + 2b^3 = 9abc \quad \checkmark$$



POLYNOMIALS

PART D: MULTIPLICITY



D1	DIFFICULTY: ☆	☒	☐
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The equation $8x^4 - 44x^3 + 54x^2 - 25x + 4 = 0$ has a triple root. Solve this equation.

3

$$\begin{aligned} P(x) &= 8x^4 - 44x^3 + 54x^2 - 25x + 4 \\ P'(x) &= 32x^3 - 132x^2 + 108x - 25 \\ P''(x) &= 96x^2 - 264x + 108 \\ &= 12(8x^2 - 22x + 9) \\ &= 12(2x - 1)(4x - 9) \end{aligned}$$

$$P\left(\frac{1}{2}\right) = P'\left(\frac{1}{2}\right) = P''\left(\frac{1}{2}\right) = 0$$

Thus $\frac{1}{2}$ is the triple root

$$\begin{aligned} 8x^4 - 44x^3 + 54x^2 - 25x + 4 &= 0 \\ (2x - 1)^3(x - 4) &= 0 \end{aligned}$$

Triple root is either $x = \frac{1}{2}$ or $x = \frac{9}{4}$

Since 4 is NOT divisible by 9, $\frac{1}{2}$ is the logical choice for triple root

$$x = \frac{1}{2} \text{ or } x = 4$$

D2	DIFFICULTY: ☆	☒	☐
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The equation $4x^3 - 27x + k = 0$ has a double root. Find the possible values of k .

3

b) Let $P(x) = 4x^3 - 27x + k = 0$

If $P(x) = 0$ has a double root
 then $P'(x) = 12x^2 - 27 = 0$ has a single root } in common

$$\begin{aligned} P'(x) &= 12x^2 - 27 = 0 \\ &= 3(4x^2 - 9) \\ &= 3(2x - 3)(2x + 3) \\ &= 0 \text{ when } x = \frac{3}{2}, -\frac{3}{2} \end{aligned}$$

If $x = \frac{3}{2}$, then $4\left(\frac{27}{8}\right) - 27\left(\frac{3}{2}\right) + k = 0$

$$\therefore k = 27$$

If $x = -\frac{3}{2}$, then $4\left(-\frac{27}{8}\right) - 27\left(-\frac{3}{2}\right) + k = 0$

$$k = -27$$

$\therefore$ The two possible values are $k = 27, -27$

D3	DIFFICULTY: ☆☆	☒	☐
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Find the multiplicity of the root $x = -1$ of the equation $P(x) = 0$ where

3

$$P(x) = x^4 + 4x^3 + 7x^2 + 6x + 2.$$

$$P(x) = x^4 + 4x^3 + 7x^2 + 6x + 2$$

$$\therefore P(-1) = -1^4 + 4(-1)^3 + 7(-1)^2 + 6(-1) + 2 = 0$$

therefore $(x + 1)$ is a factor

$$\therefore P'(x) = 4x^3 + 12x^2 + 14x + 6$$

$$\therefore P'(-1) = 4(-1)^3 + 12(-1)^2 + 14(-1) + 6 = 0$$

therefore $(x + 1)^2$ is a factor

$$\therefore P''(x) = 12x^2 + 24x + 14$$

$$\therefore P''(-1) = 12(-1)^2 + 24(-1) + 14 \neq 0$$

therefore multiplicity is 2

D4	DIFFICULTY: ☆☆	☒	☐
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$$\text{Let } P(x) = x^4 - 5x^3 + 9x^2 - 7x + 2.$$

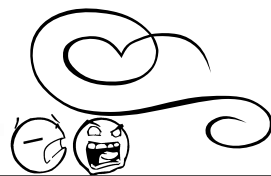
- (i) Given that $x = 1$ is a multiple root of the equation $P(x) = 0$, find the multiplicity of this root. You must justify your answer with relevant mathematical reasoning.
- (ii) Hence or otherwise factorise $P(x)$.

2

2

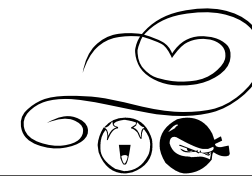
(i) $P(x) = x^4 - 5x^3 + 9x^2 - 7x + 2 \rightarrow P(1) = 0$
 $P'(x) = 4x^3 - 15x^2 + 18x - 7 \rightarrow P'(1) = 0$
 $P''(x) = 12x^2 - 30x + 18 \rightarrow P''(1) = 0$
 $P'''(x) = 24x - 30 \rightarrow P'''(1) = -6 \neq 0$
 Therefore $x = 1$ is a root of multiplicity 3.

(ii) $\therefore (x - 1)^3$ is a factor of $P(x)$
 $\therefore P(x) = (x - 1)^3(ax + b)$
 $P(0) = (0 - 1)^3(a(0) + b) = -b$,
 but $P(0) = 2$, $\therefore -b = 2 \rightarrow b = -2$
 Equating the coefficient of x^4 .
 We get $ax^4 = x^4 \rightarrow a = 1$
 $\therefore P(x) = (x - 1)^3(x - 2)$.



POLYNOMIALS

PART D: MULTIPLICITY



D5	DIFFICULTY: ☆	☒	☐
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$P(x) = x^4 + 7x^3 + 9x^2 - 27x + c$ has a zero of multiplicity 3. Find the value of c . 3

$$P'(x) = 4x^3 + 21x^2 + 18x - 27$$

$$P''(x) = 12x^2 + 42x + 18$$

$$P''(x) = 0 \quad x = -\frac{1}{2}$$

$$2x^2 + 7x + 3 = 0$$

D6	DIFFICULTY: ☆☆	☒	☐
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The polynomial $P(x) = ax^4 + bx^3 + cx^2 + e$ has a remainder -3 when divided by $x - 1$. 3
The polynomial has a double root at $x = -1$. Find the value of $4a + 2c$.

$$P(x) = ax^4 + bx^3 + cx^2 + e$$

$$P'(x) = 4ax^3 + 3bx^2 + 2cx$$

$$\text{①} - \text{②} \quad 2b = -3 \quad b = -\frac{3}{2}$$

$$P(1) = a + b + c + e = -3 \quad \text{--- ①} \quad \text{substitute } b = -\frac{3}{2} \text{ into ③}$$

$$P(-1) = a - b + c + e = 0 \quad \text{--- ②}$$

$$P'(-1) = -4a + 3b - 2c = 0 \quad 4a + 2c = 3(-\frac{3}{2})$$

$$\therefore 4a + 2c = 3b - \text{③} \quad \therefore 4a + 2c = -\frac{9}{2}$$

D7	DIFFICULTY: ☆☆	☒	☐
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The polynomial $P(x) = 2x^3 + mx^2 + nx - 20$ has a double root and $P(2) = P'(2) = 0$. 3
Find the values of m and n and find the other root of $P(x)$.

$P(x) = 2x^3 + mx^2 + nx - 20$ has a double root,
so let roots be α, β, β

Since $P(2) = P'(2) = 0$,
the double root is $x = \beta = 2$

$$P(2) = 16 + 4m + 2n - 20 = 0$$

$$4m + 2n = 4 \quad \dots \text{①}$$

$$P'(2) = 24 + 4m + n = 0$$

$$4m + n = -24 \quad \dots \text{②}$$

$$2n - n = 4 - 24 \quad \dots \text{①} - \text{②}$$

$$n = -20$$

$$4m + 2(-20) = 4$$

$$4m + 56 = 4$$

$$4m = -52$$

$$m = \frac{-52}{4} = -13$$

Roots of $P(x) = 2x^3 - 13x^2 + 28x - 20$
 $= (x - 2)^2(2x - \alpha) = 0$ can then be found

by inspection or as follows:

For $ax^3 + bx^2 + cx + d$ with roots $\alpha, 2$ and 2

$$\alpha + 2 + 2 = -\frac{b}{a} \Rightarrow \alpha + 4 = -\frac{(-13)}{2}$$

$$2\alpha + 8 = 13$$

$$2\alpha = 5$$

$$\alpha = \frac{5}{2}$$

So $m = -13$ and $n = 28$, and the roots are

$\alpha = \frac{5}{2}$ and a double root $\beta = 2$

D8	DIFFICULTY: ☆☆	☒	☐
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A polynomial function $y = f(x)$ has a zero of multiplicity 3 at $x = 0$ and a zero of multiplicity 1 at $x = -3$. The polynomial is an odd function and its graph passes through the point $(-2, -20)$. Write down the equation of the polynomial, of least degree, which satisfies the given description. 3

Since $x = 0$ is a triple zero and $x = -3$ is a single zero, then $(x - 0)^3$ and $(x + 3)$ are factors of $y = f(x)$, by the Factor Theorem.

Also, since the function is odd, it has point symmetry about the origin. Hence, since $x = -3$ is a zero, then $x = 3$ is also a zero. $\therefore (x - 3)$ is a factor of $f(x)$, by the Factor Theorem.

$\therefore$ the required equation is of the form:
 $y = ax^3(x + 3)(x - 3)$

Since the curve passes through the point $(-2, -20)$:

$$-20 = a(-2)^3(-2 + 3)(-2 - 3)$$

$$-20 = 40a$$

$$a = \frac{-20}{40} = -\frac{1}{2}$$

$$\therefore y = -\frac{1}{2}x^3(x + 3)(x - 3)$$

D9	DIFFICULTY: ☆☆	☒	☐
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Let $P(x) = x^3 + ax^2 + bx + 5$ where a and b are real numbers. 2
Find the values of a and b given that $(x - 1)^2$ is a factor of $P(x)$.

$$P(x) = x^3 + ax^2 + bx + 5$$

$$P'(x) = 3x^2 + 2ax + b$$

Since $(x - 1)^2$ is a factor of $P(x)$ then $P(1) = 0$
 $P'(1) = 0$

$$P(1) = 1 + a + b + 5 = 0$$

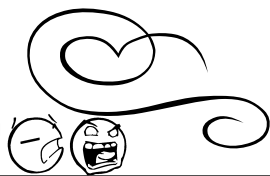
$$a + b = -6$$

$$P'(1) = 3 + 2a + b = 0$$

$$2a + b = -3$$

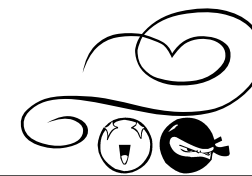
$$a = 3$$

$$b = -9$$



POLYNOMIALS

Part D: Multiplicity



D10	DIFFICULTY: ☆☆	☒	☐
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The polynomial $P(x) = x^4 - 10x^3 + 24x^2 + 32x + k$ has a triple zero.

- (i) Determine the value of the triple zero.
- (ii) Hence find the value of k .
- (iii) Fully factorise $P(x)$.

**3
1
2**

(b) $P(x) = x^4 - 10x^3 + 24x^2 + 32x + k$

(i) $P'(x) = 4x^3 - 30x^2 + 48x + 32$
 $P''(x) = 12x^2 - 60x + 48$
 Let $P''(x) = 0$: $12x^2 - 60x + 48 = 0$ ✓
 $x^2 - 5x + 4 = 0$
 $(x-1)(x-4) = 0$
 $\therefore$ the triple zero is either 1 or 4. ✓

$P'(1) = 4 - 30 + 48 + 32 = 54 \neq 0$

Since it is given that $P(x)$ has a triple zero, it can be deduced that it is 4. ✓

(ii) $P(4) = 0$: $4^4 - 10 \times 4^3 + 24 \times 4^2 + 32 \times 4 + k = 0$
 $128 + k = 0$
 $k = -128$ ✓

(iii) Let the zeros be 4, 4, 4, α .
 Sum of zeros: $4 + 4 + 4 + \alpha = -\frac{-10}{1}$
 $\alpha + 12 = 10$
 $\alpha = -2$ ✓
 $\therefore P(x) = (x-4)^3(x+2)$ ✓

D11	DIFFICULTY: ☆	☒	☐
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Where does the polynomial $P(x) = 4x^3 - 8x^2 - 11x - 3$ have a root of multiplicity 2? **1**

A. $x = -\frac{1}{2}$

C. $x = \frac{11}{6}$

B. $x = \frac{2}{3}$

D. $x = 3$

Given $P(x) = 4x^3 - 8x^2 - 11x - 3$,
 $P'(x) = 12x^2 - 16x - 11$, and
 $P''(x) = 24x - 16$

Option A satisfies $P(x)$ and $P'(x)$,
 hence it is a root of multiplicity 2.

D12	DIFFICULTY: ☆☆	☒	☐
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Show that, if $x^3 + px + r = 0$ has a root of multiplicity two, then $27r^2 + 4p^3 = 0$ **2**

$x^3 + px + r = 0$

Let the roots be α, α, β

$\therefore 2\alpha + \beta = 0$

$\beta = -2\alpha$

$\alpha^2 + 2\alpha\beta = p$

$\therefore p = \alpha^2 + 2\alpha(-2\alpha) = -3\alpha^2$

and $\alpha^2\beta = -r$

$-r = \alpha^2(-2\alpha) \therefore r = 2\alpha^3$

$\therefore 27r^2 + 4p^3 = 27(4\alpha^6) + 4(-27\alpha^6) = 0$

D13	DIFFICULTY: ☆☆	☒	☐
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Let $P(x) = 2x^4 - 15x^3 + 2x^2 + ax + b$, where a and b are real numbers. **3**

Find the values of a and b given that $(x-5)^2$ is a factor of $P(x)$.

$P(x) = 2x^4 - 15x^3 + 2x^2 + ax + b$

$P'(x) = 8x^3 - 45x^2 + 4x + a$

$P(5) = 2(5)^4 - 15(5)^3 + 2(5)^2 + 5a + b$

$P'(5) = 8(5)^3 - 45(5)^2 + 4(5) + a$

$= 0$

$= 0$

$-575 + 5a + b = 0$

$-105 + a = 0$

$5a + b = 575$ (1)

$a = 105$ (2)

Substituting (2) into (1) gives:

$5(105) + b = 575$

$b = 50$